LESSON PLAN

WEAPON GUIDANCE (19)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	11
Total No. of Lesson Plans	22	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Explanation on clamping and bend radius of wire and cables
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Explanation on clamping and bend radius of wire and cables.
5. Specific Objectives	19.1.1.1Trainee will be able to explain aircraft wiring and Causes of wiring degradation.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Demonstration of proper clamping techniques and measurement of minimum bend radius using sample wire bundles; Safety: Ensure no loose wires cause short circuits; Supervision: Instructor monitors tool usage and measurement accuracy.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What are the primary causes of aircraft wiring degradation?
(b) Why is maintaining the correct bend radius critical for wire integrity?
(c) How does improper clamping affect the electrical performance of a wire bundle?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Definition of minimum bend radius • Consequences of exceeding bend radius • Standard bend radius ratios for different wire types • Impact of sharp bends on insulation and conductors • Visual identification of damaged wires due to bending	L & Ds	PPT, IB	15 min
(b)	• Types of clamps used in aircraft wiring • Proper installation techniques for wire clamps • Common errors in clamp installation • Effect of loose clamps on wire bundle stability • Inspection criteria for clamp security and alignment	L & Ds	PPT, PDF	20 min
(c)	• Relationship between clamping and bend radius • Preventing stress points at clamp locations • Best practices for routing wires near clamps • Maintenance checks for clamp-induced degradation • Case studies of wiring failures due to poor clamping	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how improper clamping can lead to wire degradation.
(b) Describe the minimum bend radius requirement for a standard aircraft wire.
(c) Identify three signs of wiring damage caused by incorrect bending or clamping.

12. Home Assignment and Follow Up	Prepare a brief report outlining the steps to inspect a wire bundle for clamp-related degradation, referencing the guidelines in LP 12 regarding wire bundle breaks.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Explanation on various types of wire bundle breaks and their uses.	Refer to the master syllabus reference list	LP 12

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min